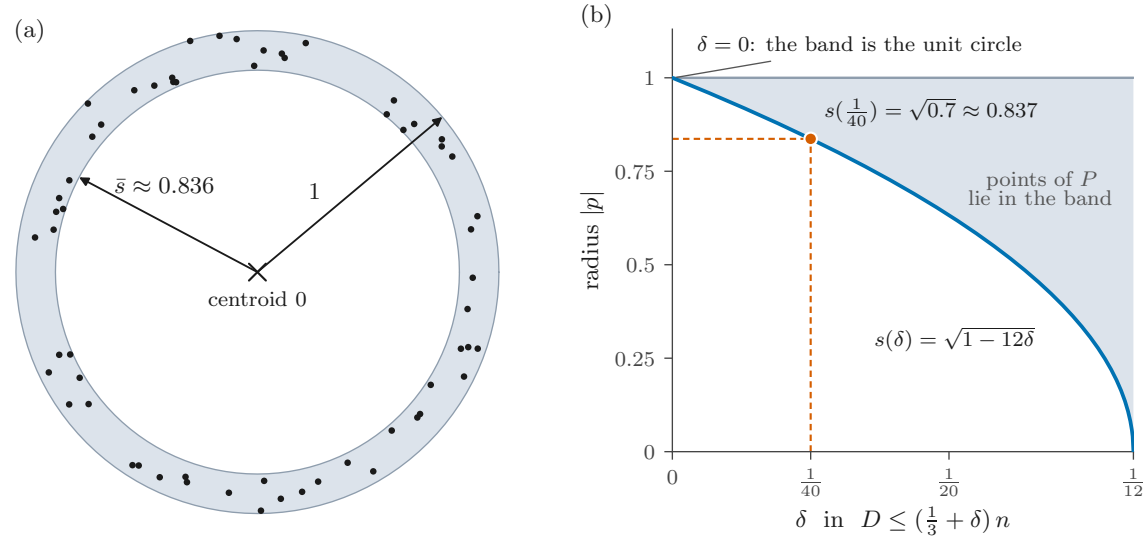

(a)
$\bar{s} \approx 0.836$
1
centroid 0
(b)
$\delta = 0$: the band is the unit circle
$s(\frac{1}{40}) = \sqrt{0.7} \approx 0.837$
points of P lie in the band
$s(\delta) = \sqrt{1 - 12\delta}$
radius $|p|$
1
0.75
0.5
0.25
0
0
$\frac{1}{40}$
$\frac{1}{20}$
$\frac{1}{12}$
δ in $D \leq (\frac{1}{3} + \delta)\, n$